Lecture 8: Asymptotic theory

Economics 527: Econometric Methods

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Convergence in probability

OLS as a function of sample averages

- Model and OLS estimator:

$$Y_i = X_i^\top \beta + U_i, \qquad \hat{\beta} = \Big(\sum_{i=1}^n X_i X_i^\top \Big)^{-1} \sum_{i=1}^n X_i Y_i$$

- Normal errors, exact distribution given the regressors:

$$\begin{aligned} & U_i \mid X_i \sim N(0, \sigma^2) \\ \implies \quad & \hat{\beta} \mid X_1, \dots, X_n \sim N\Big(\beta,\ \sigma^2 \Big(\sum_{i=1}^n X_i X_i^\top \Big)^{-1} \Big) \end{aligned}$$

- Without normality, write OLS in terms of sample averages:

$$\hat{\beta} = \Big(\frac{1}{n} \sum_{i=1}^n X_i X_i^\top \Big)^{-1} \frac{1}{n} \sum_{i=1}^n X_i Y_i$$

- Use approximations; all limits below are as $n \to \infty$.

Limit of a non-random sequence

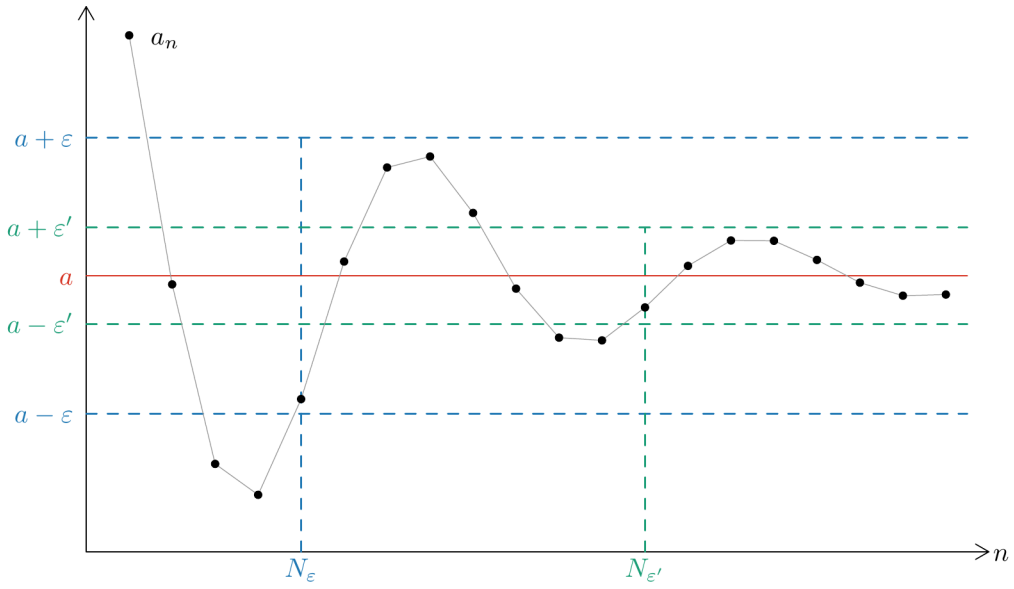
- Non-random sequence:

$$\{ \underset{1\times 1}{a_n} \in \mathbb{R} : n \geq 1 \}$$

- **Definition.** a_n converges to a, $a_n \to a$, $a = \lim a_n$, if for all $\varepsilon > 0$ there is $N_\varepsilon \in \mathbb{N}$:

$$|a_n - a| < \varepsilon \quad \text{for all } n \geq N_\varepsilon$$

- a: good approximation of a_n for all $n \geq N_\varepsilon$.
- Sequence a_n with bands $a \pm \varepsilon$ and $a \pm \varepsilon'$, $\varepsilon' < \varepsilon$:



Estimator as a random sequence

- $\hat{\theta}_n$: estimator of θ from n observations; random sequence:

$$\{\underset{1\times 1}{\hat{\theta}_n} : n \geq 1\}, \qquad \hat{\theta}_n = \hat{\theta}_n(\omega), \quad \omega \in \Omega$$

- OLS from n observations: $\hat{\beta}_n = \hat{\beta}_n(\omega)$, random.
- Non-random sequences from $\hat{\theta}_n$; apply the definition of a limit to them:

$$\begin{aligned} \text{Moments, } r > 0: &\quad a_n = \mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] \\ \text{Probabilities, } \varepsilon > 0: &\quad a_n = \mathrm{P}(|\hat{\theta}_n - \theta| > \varepsilon) \end{aligned}$$

Convergence in r-th mean

- **Definition.** $\hat{\theta}_n$ converges to θ in r-th mean, $r > 0$, if:

$$\mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] \to 0$$

- $a_n = \mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] \to 0$: for all $\varepsilon > 0$ there is $N_\varepsilon \in \mathbb{N}$:

$$\mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] < \varepsilon \quad \text{for all } n \geq N_\varepsilon$$

- $r = 2$, mean squared error (MSE):

$$\mathrm{E}\left[(\hat{\theta}_n - \theta)^2\right]$$

Definition of convergence in probability

- Non-random sequence, $\delta > 0$ fixed:

$$a_n = \mathrm{P}(|\hat{\theta}_n - \theta| > \delta), \qquad n \geq 1$$

- **Definition.** $\hat{\theta}_n$ converges in probability to θ, $\hat{\theta}_n \to_p \theta$, if for all $\delta > 0$:

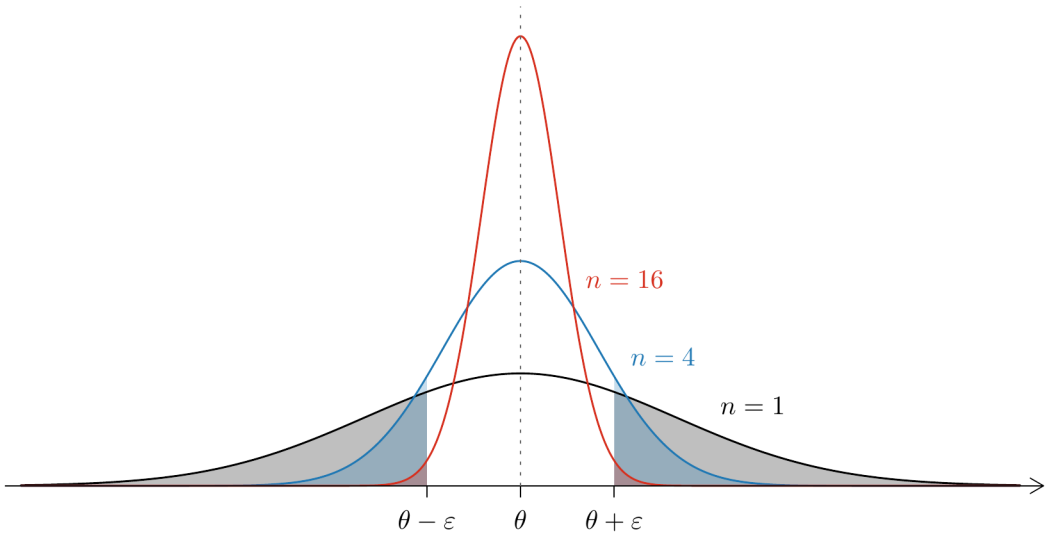
$$\mathrm{P}(|\hat{\theta}_n - \theta| > \delta) \to 0$$

- Equivalently, for all $\delta > 0$:

$$\begin{aligned} \mathrm{P}(|\hat{\theta}_n - \theta| \leq \delta) &= 1 - \underbrace{\mathrm{P}(|\hat{\theta}_n - \theta| > \delta)}_{\to 0} \\ &\to 1 \end{aligned}$$

Densities of $\hat{\theta}_n$ as n grows

- Example: densities for $\hat{\theta}_n \sim N(\theta, 1/n)$.
- Shaded: $\mathrm{P}(|\hat{\theta}_n - \theta| > \varepsilon)$.



- $\hat{\theta}_n \to_p \theta$: distribution of $\hat{\theta}_n$ more concentrated around θ.

Markov's inequality

- Markov's inequality connects tail probabilities to expectations; the core device for establishing convergence in probability.
- **Theorem.** X: random variable, $M > 0$:

$$\mathrm{P}(|X| \geq M) \leq \frac{\mathrm{E}\left[|X|\right]}{M}$$

- Expectation of an indicator, X with density f_X, split at $-M$ and M:

$$\begin{aligned}
\mathrm{E}\left[\mathbf{1}(|X| \geq M)\right] &= \int_{-\infty}^{\infty} \mathbf{1}(|x| \geq M)\, f_X(x)\, dx \\
&= \int_{-\infty}^{-M} \underbrace{\mathbf{1}(|x| \geq M)}_{=1} f_X(x)\, dx \\
&\quad + \overbrace{\int_{-M}^{M} \underbrace{\mathbf{1}(|x| \geq M)}_{=0 \text{ for } |x|<M} f_X(x)\, dx}^{=0} \\
&\quad + \int_{M}^{\infty} \underbrace{\mathbf{1}(|x| \geq M)}_{=1} f_X(x)\, dx \\
&= \int_{-\infty}^{-M} f_X(x)\, dx + \int_{M}^{\infty} f_X(x)\, dx \\
&= \mathrm{P}(X \leq -M) + \mathrm{P}(X \geq M) \\
&= \mathrm{P}(|X| \geq M)
\end{aligned}$$

- Split 1 into two indicators:

$$1 = \mathbf{1}(|X| \geq M) + \mathbf{1}(|X| < M)$$

- Expectation of $|X|$:

$$\begin{aligned}
\mathrm{E}\left[|X|\right] &= \mathrm{E}\left[|X| \cdot 1\right] \\
&= \mathrm{E}\left[|X| \left(\mathbf{1}(|X| \geq M) + \mathbf{1}(|X| < M)\right)\right] \\
&= \mathrm{E}\left[|X|\, \mathbf{1}(|X| \geq M)\right] + \underbrace{\mathrm{E}\left[|X|\, \mathbf{1}(|X| < M)\right]}_{\geq 0} \\
&\geq \mathrm{E}\left[|X|\, \mathbf{1}(|X| \geq M)\right] \\
&\geq \mathrm{E}\left[M\, \mathbf{1}(|X| \geq M)\right] \quad \substack{|X| \geq M \\ \text{where } \mathbf{1}(|X| \geq M)=1} \\
&= M\, \mathrm{E}\left[\mathbf{1}(|X| \geq M)\right] \\
&= M\, \mathrm{P}(|X| \geq M)
\end{aligned}$$

- Divide by M:

$$\boxed{\mathrm{P}(|X| \geq M) \leq \frac{\mathrm{E}\left[|X|\right]}{M}}$$

Convergence in r-th mean and in probability

- **Claim.**

$$\mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] \to 0 \;\text{ for some } r > 0 \quad \Longrightarrow \quad \hat{\theta}_n \to_p \theta$$

- Markov's inequality, for all $\varepsilon > 0$:

$$\begin{aligned}
\mathrm{P}(|\hat{\theta}_n - \theta| > \varepsilon) &= \mathrm{P}(|\hat{\theta}_n - \theta|^r > \varepsilon^r) \\
&\leq \mathrm{P}(\underbrace{|\hat{\theta}_n - \theta|^r}_{X} \geq \underbrace{\varepsilon^r}_{M}) \\
&\leq \frac{1}{\varepsilon^r} \underbrace{\mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right]}_{\to 0} \\
&\to 0
\end{aligned}$$

- Therefore:

$$\boxed{\hat{\theta}_n \to_p \theta}$$

Chebyshev's inequality

- As on the previous slide, $r = 2$, $Y - \mathrm{E}[Y]$ in place of $\hat{\theta}_n - \theta$, $\varepsilon > 0$:

$$\begin{aligned}\mathrm{P}(|Y - \mathrm{E}[Y]| > \varepsilon) &\leq \frac{1}{\varepsilon^2}\mathrm{E}\left[|Y - \mathrm{E}[Y]|^2\right] \\ &= \frac{1}{\varepsilon^2}\mathrm{Var}(Y)\end{aligned}$$

Bias and variance tending to zero

- **Claim.** For an estimator $\hat{\theta}_n$ of θ:

$$\begin{aligned}&1.\ \mathrm{E}\left[\hat{\theta}_n\right] \to \theta \\ &2.\ \mathrm{Var}\left(\hat{\theta}_n\right) \to 0 \\ \implies\ &\hat{\theta}_n \to_p \theta\end{aligned}$$

- Recall:

$$\underbrace{\mathrm{E}\left[(\hat{\theta}_n - \theta)^2\right]}_{\text{MSE}} = \mathrm{Var}\left(\hat{\theta}_n\right) + \underbrace{\left(\mathrm{E}\left[\hat{\theta}_n\right] - \theta\right)^2}_{\text{Bias}^2}$$

- Markov with $r = 2$, for all $\varepsilon > 0$:

$$\begin{aligned}\mathrm{P}(|\hat{\theta}_n - \theta| > \varepsilon) &\leq \frac{1}{\varepsilon^2}\mathrm{E}\left[(\hat{\theta}_n - \theta)^2\right] \\ &= \frac{1}{\varepsilon^2}\Big(\underbrace{\mathrm{Var}\left(\hat{\theta}_n\right)}_{\to 0 \text{ by 2.}} + (\underbrace{\mathrm{E}\left[\hat{\theta}_n\right] - \theta}_{\to 0 \text{ by 1.}})^2\Big) \\ &\to 0\end{aligned}$$

- Therefore:

$$\boxed{\hat{\theta}_n \to_p \theta}$$

Law of large numbers (LLN)

LLN for uncorrelated observations

- **Theorem.** $X_1, \ldots, X_n$: random variables; for all i and all $j \neq i$:

$$\begin{aligned}&\mathrm{E}[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2 < \infty, \qquad \mathrm{Cov}(X_i, X_j) = 0 \\ \implies\ &\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \to_p \mu\end{aligned}$$

Proof of the LLN, step 1: mean and variance

- $\mathrm{E}\left[\bar{X}_n\right] = \mu$, so bias is zero:

$$\begin{aligned}\mathrm{E}\left[\bar{X}_n\right] &= \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}[X_i] \\ &= \mu\end{aligned}$$

- Variance of $\bar{X}_n$:

$$\begin{aligned}
\operatorname{Var}\left(\bar{X}_n\right) &= \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\operatorname{Cov}\left(X_i, X_j\right) \\
&= \frac{1}{n^2}\Big(\sum_{i=1}^{n}\operatorname{Var}\left(X_i\right) + \underbrace{\sum_{i=1}^{n}\sum_{j\neq i}\operatorname{Cov}\left(X_i, X_j\right)}_{=0}\Big) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\underbrace{\operatorname{Var}\left(X_i\right)}_{=\sigma^2} \\
&= \frac{1}{n^2}\left(\sigma^2 \cdot n\right) \\
&= \frac{\sigma^2}{n} \\
&\to 0
\end{aligned}$$

Proof of the LLN, step 2: Chebyshev's inequality

- Chebyshev with $\mathrm{E}\left[\bar{X}_n\right] = \mu$, for all $\varepsilon > 0$:

$$\begin{aligned}
\mathrm{P}(|\bar{X}_n - \mu| > \varepsilon) &\leq \frac{1}{\varepsilon^2}\operatorname{Var}\left(\bar{X}_n - \mu\right) \\
&= \frac{1}{\varepsilon^2}\operatorname{Var}\left(\bar{X}_n\right) \\
&= \frac{1}{\varepsilon^2}\frac{\sigma^2}{n} \\
&\to 0
\end{aligned}$$

- Therefore:

$$\boxed{\bar{X}_n \to_p \mu = \mathrm{E}\left[X_i\right]}$$

Extension to different means

- $\mathrm{E}\left[X_i\right] = \mu_i$, not constant; $\operatorname{Var}\left(X_i\right) = \sigma^2 < \infty$; $\operatorname{Cov}\left(X_i, X_j\right) = 0$ for $i \neq j$.
- Centre each X_i at its mean:

$$\begin{aligned}
& Y_i = X_i - \mathrm{E}\left[X_i\right] \\
\implies \quad & \mathrm{E}\left[Y_i\right] = 0, \qquad \operatorname{Var}\left(Y_i\right) = \sigma^2, \qquad \operatorname{Cov}\left(Y_i, Y_j\right) = 0 \ \text{ for } i \neq j
\end{aligned}$$

- LLN for $Y_1, \ldots, Y_n$:

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}\left[X_i\right] &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \mathrm{E}\left[X_i\right]\right) \\
&= \frac{1}{n}\sum_{i=1}^{n} Y_i \\
&\to_p 0
\end{aligned}$$

Extension to different variances

- $Y_1, \ldots, Y_n$: random variables; for all i and all $j \neq i$:

$$\begin{aligned} &\mathrm{E}\left[Y_i\right] = 0, \qquad \mathrm{Var}\left(Y_i\right) = \mathrm{E}\left[Y_i^2\right] = \sigma_i^2 \quad \text{not constant} \\ &\mathrm{Cov}\left(Y_i, Y_j\right) = 0 \end{aligned}$$

- Bounded variances, $\bar{\sigma}^2$ not depending on n:

$$\sigma_i^2 \leq \max_{j=1,\ldots,n} \sigma_j^2 \leq \bar{\sigma}^2$$

- Variance of the average $\to 0 \implies \frac{1}{n}\sum_{i=1}^n Y_i \to_p 0$, by Chebyshev:

$$\begin{aligned} \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n Y_i\right) &= \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^n Y_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}\left(Y_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^n \sigma_i^2 \\ &\leq \frac{1}{n^2} n\bar{\sigma}^2 \\ &= \frac{\bar{\sigma}^2}{n} \\ &\to 0 \end{aligned}$$

Extension to dependent observations

- Variance of the average, non-zero covariances:

$$\begin{aligned} \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n Y_i\right) &= \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n \mathrm{Cov}\left(Y_i, Y_j\right) \\ &= \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}\left(Y_i\right) + \frac{1}{n^2}\sum_{i=1}^n\sum_{j\neq i} \mathrm{Cov}\left(Y_i, Y_j\right) \\ &= \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}\left(Y_i\right) + 2\,\frac{1}{n^2}\sum_{i=1}^{n-1}\sum_{j=i+1}^n \mathrm{Cov}\left(Y_i, Y_j\right) \end{aligned}$$

- Number of distinct covariances, order n^2:

$$\frac{n^2 - n}{2} = \frac{n(n-1)}{2}$$

- Triangle inequality, $|a+b| \leq |a| + |b|$:

$$\begin{aligned} \left|\frac{1}{n^2}\sum_{i=1}^n\sum_{j\neq i} \mathrm{Cov}\left(Y_i, Y_j\right)\right| &\leq \frac{1}{n^2}\underbrace{\sum_{i=1}^n\sum_{j\neq i} \left|\mathrm{Cov}\left(Y_i, Y_j\right)\right|}_{\text{if of smaller order than } n^2} \\ &\to 0 \end{aligned}$$

- Double sum smaller order than n^2: for example, each observation correlated with only a fixed number of others, with bounded covariances.

Example: $Y_i = \varepsilon_i + \varepsilon_{i+1}$

- $\varepsilon_1, \varepsilon_2, \ldots$: iid (independent and identically distributed), $0 < \operatorname{Var}(\varepsilon_i) < \infty$.
- Neighbours share one ε:

$$\begin{aligned} Y_i &= \varepsilon_i + \varepsilon_{i+1} \\ Y_{i+1} &= \varepsilon_{i+1} + \varepsilon_{i+2} \\ Y_{i+2} &= \varepsilon_{i+2} + \varepsilon_{i+3} \end{aligned}$$

- Covariances of Y_i and Y_{i+h}:

$$\operatorname{Cov}(Y_i, Y_{i+1}) \neq 0, \qquad \operatorname{Cov}(Y_i, Y_{i+h}) = 0 \quad \text{for } h \geq 2$$

Weak LLN for iid observations

- Independence instead of uncorrelatedness: finite variance not required.
- **Theorem.** $X_1, \ldots, X_n$: iid random variables:

$$\begin{aligned} & \operatorname{E}[|X_1|] < \infty \\ \Longrightarrow \quad & \bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \to_p \operatorname{E}[X_1] \\ & \operatorname{E}[X_1] = \int_{-\infty}^{\infty} x \, f_X(x) \, dx \quad \text{with density } f_X \end{aligned}$$

- Stated without proof; this is the LLN we will use.

Vectors and Slutsky's theorem

Convergence in probability of a random vector

- Estimator $\hat{\theta}_n$ and parameter θ, $k \times 1$:

$$\hat{\theta}_n = \begin{pmatrix} \hat{\theta}_{n,1} \\ \vdots \\ \hat{\theta}_{n,k} \end{pmatrix}, \qquad \theta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_k \end{pmatrix}$$

- **Definition.** $\hat{\theta}_n$ converges in probability to θ, $\hat{\theta}_n \to_p \theta$, two versions:

1. Element by element: $\hat{\theta}_{n,j} \to_p \theta_j$ for all $j = 1, \ldots, k$
2. In norm: $\|\hat{\theta}_n - \theta\| \to_p 0, \qquad \|x\| = \sqrt{x_1^2 + \cdots + x_k^2}$

- **Claim.** $1 \iff 2$.

$$\|\hat{\theta}_n - \theta\|^2 = (\hat{\theta}_{n,1} - \theta_1)^2 + \cdots + (\hat{\theta}_{n,k} - \theta_k)^2$$

- $1 \Longrightarrow 2$: each element converges to zero, so the finite sum converges to zero.
- $2 \Longrightarrow 1$: non-negative summands; the sum converges to zero only if each summand does.

Sum of two sequences

- **Claim.** For 1×1 sequences $\hat{\theta}_{n,1}$ and $\hat{\theta}_{n,2}$:

$$\begin{aligned} & \hat{\theta}_{n,1} \to_p \theta_1, \quad \hat{\theta}_{n,2} \to_p \theta_2 \\ \implies \quad & \hat{\theta}_{n,1} + \hat{\theta}_{n,2} \to_p \theta_1 + \theta_2 \end{aligned}$$

- Triangle inequality, then a union bound, for all $\varepsilon > 0$:

$$\begin{aligned} & \mathrm{P}(|\hat{\theta}_{n,1} + \hat{\theta}_{n,2} - (\theta_1 + \theta_2)| > \varepsilon) \\ & \leq \mathrm{P}(|\hat{\theta}_{n,1} - \theta_1| + |\hat{\theta}_{n,2} - \theta_2| > \varepsilon) \\ & \leq \mathrm{P}\Big(|\hat{\theta}_{n,1} - \theta_1| > \frac{\varepsilon}{2} \ \text{ or } \ |\hat{\theta}_{n,2} - \theta_2| > \frac{\varepsilon}{2}\Big) \\ & \leq \mathrm{P}\Big(|\hat{\theta}_{n,1} - \theta_1| > \frac{\varepsilon}{2}\Big) + \mathrm{P}\Big(|\hat{\theta}_{n,2} - \theta_2| > \frac{\varepsilon}{2}\Big) \\ & \to 0 \end{aligned}$$

- Therefore:

$$\boxed{\hat{\theta}_{n,1} + \hat{\theta}_{n,2} \to_p \theta_1 + \theta_2}$$

Slutsky's theorem

- More general than sums: convergence is preserved by continuous functions.
- **Theorem.** $\hat{\theta}_n$: $k \times 1$; $\theta \in \mathbb{R}^k$ non-random; $h : \mathbb{R}^k \to \mathbb{R}^m$ continuous at θ; $\to_p$ in norm:

$$\hat{\theta}_n \to_p \theta \quad \implies \quad h(\hat{\theta}_n) \to_p h(\theta)$$

- h continuous at θ: for all $\varepsilon > 0$ there is $\delta_\varepsilon > 0$:

$$\|\hat{\theta}_n - \theta\| \leq \delta_\varepsilon \quad \implies \quad \|h(\hat{\theta}_n) - h(\theta)\| \leq \varepsilon$$

- Inclusion of events, then probabilities, for all $\varepsilon > 0$:

$$\begin{aligned} & \{\|\hat{\theta}_n - \theta\| \leq \delta_\varepsilon\} \subset \{\|h(\hat{\theta}_n) - h(\theta)\| \leq \varepsilon\} \\ \implies \quad & \underbrace{\mathrm{P}(\|\hat{\theta}_n - \theta\| \leq \delta_\varepsilon)}_{\to 1 \text{ (in norm)}} \leq \mathrm{P}(\|h(\hat{\theta}_n) - h(\theta)\| \leq \varepsilon) \leq 1 \\ \implies \quad & \mathrm{P}(\|h(\hat{\theta}_n) - h(\theta)\| \leq \varepsilon) \to 1 \\ \implies \quad & \boxed{h(\hat{\theta}_n) \to_p h(\theta)} \end{aligned}$$

Examples: product, ratio

- $\hat{\theta}_{n,1} \to_p \theta_1$, $\hat{\theta}_{n,2} \to_p \theta_2$; element by element $\implies$ in norm; Slutsky for product and ratio:

$$\begin{aligned} & \hat{\theta}_{n,1}\, \hat{\theta}_{n,2} \to_p \theta_1\, \theta_2 \\ & \frac{\hat{\theta}_{n,1}}{\hat{\theta}_{n,2}} \to_p \frac{\theta_1}{\theta_2} \quad \text{if } \theta_2 \neq 0 \end{aligned}$$

- $h(\theta_1, \theta_2) = \theta_1/\theta_2$: not continuous at $\theta_2 = 0$.
- $X_1, \ldots, X_n$ iid, $\mathrm{E}\,[|X_1|] < \infty$; weak LLN, then Slutsky for $h(u) = u^2$ and $h(u) = 1/u$:

$$\begin{aligned} & (\bar{X}_n)^2 \to_p \left(\mathrm{E}\,[X_1]\right)^2 \\ & \frac{1}{\bar{X}_n} \to_p \frac{1}{\mathrm{E}\,[X_1]} \quad \text{if } \mathrm{E}\,[X_1] \neq 0 \end{aligned}$$

Example: inverse of a matrix

- X_i: $k \times 1$, iid, $\mathrm{E}\left[X_i X_i^\top\right]$ finite; weak LLN, element by element:

$$\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top \to_p \mathrm{E}\left[X_i X_i^\top\right]$$

- $\mathrm{E}\left[X_i X_i^\top\right]$: symmetric, p.s.d.
- Assume $\mathrm{E}\left[X_i X_i^\top\right]$ p.d.; $h(\cdot) = (\cdot)^{-1}$: continuous at a p.d. matrix.
- Element by element $\Longrightarrow$ in norm, then Slutsky on the k^2 elements:

$$\left(\frac{1}{n}\sum_{i=1}^{n} X_i X_i^\top\right)^{-1} \to_p \left(\mathrm{E}\left[X_i X_i^\top\right]\right)^{-1}$$

Convergence in distribution

Normal distributions and Φ

- Standard normal CDF:

$$\Phi(t) = \mathrm{P}(Z \le t), \qquad Z \sim N(0,1)$$

- For $X \sim N(\mu, \sigma^2)$, $\sigma > 0$:

$$\begin{aligned}\mathrm{P}(X \le t) &= \mathrm{P}(X - \mu \le t - \mu) \\ &= \mathrm{P}\Big(\underbrace{\frac{X-\mu}{\sigma}}_{=Z\sim N(0,1)} \le \frac{t-\mu}{\sigma}\Big) \\ &= \Phi\Big(\frac{t-\mu}{\sigma}\Big)\end{aligned}$$

- For any X, $\mu \in \mathbb{R}$ and $\sigma > 0$:

$$\mathrm{P}(X \le t) = \Phi\Big(\frac{t-\mu}{\sigma}\Big) \text{ for all } t \in \mathbb{R} \quad \Longleftrightarrow \quad X \sim N(\mu, \sigma^2)$$

Asymptotic normality

- $\hat{\theta}_n$, θ: 1×1.
- **Definition.** $\hat{\theta}_n$ asymptotically normal: for some $\omega > 0$:

$$\begin{aligned}\mathrm{P}(\sqrt{n}(\hat{\theta}_n - \theta) \le t) &= \mathrm{P}\Big(\frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\omega} \le \frac{t}{\omega}\Big) \\ &\to \Phi\Big(\frac{t}{\omega}\Big) \quad \text{for all } t \in \mathbb{R} \\ \Longleftrightarrow \quad \frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\omega} &\to_d N(0,1) \\ \Longleftrightarrow \quad \sqrt{n}(\hat{\theta}_n - \theta) &\to_d N(0, \omega^2)\end{aligned}$$

- Convergence in distribution concerns distribution functions, not convergence of the random variables themselves.
- Approximate using the normal CDF; valid as $n \to \infty$:

$$\mathrm{P}(\sqrt{n}(\hat{\theta}_n - \theta) \le t) \approx \Phi\Big(\frac{t}{\omega}\Big)$$

Why $\sqrt{n}$

- $X_1, \ldots, X_n$ iid, mean θ, variance $0 < \sigma^2 < \infty$; $\hat{\theta}_n = \bar{X}_n$:

$$\hat{\theta}_n - \theta \to_p 0 \quad \text{(LLN)}$$

- Variance of $\hat{\theta}_n - \theta$ (proof of the LLN, step 1):

$$\begin{aligned} \operatorname{Var}\left(\hat{\theta}_n - \theta\right) &= \operatorname{Var}\left(\bar{X}_n\right) \\ &= \frac{\sigma^2}{n} \\ &\to 0 \end{aligned}$$

- Scaled by $\sqrt{n}$:

$$\begin{aligned} \operatorname{Var}\left(\sqrt{n}(\hat{\theta}_n - \theta)\right) &= n \operatorname{Var}\left(\hat{\theta}_n - \theta\right) \\ &= n \cdot \frac{\sigma^2}{n} \\ &= \sigma^2 \end{aligned}$$

- Scaled by n^α:

$$\begin{aligned} \operatorname{Var}\left(n^\alpha(\hat{\theta}_n - \theta)\right) &= \frac{n^{2\alpha}}{n}\,\sigma^2 \\ &= n^{2\alpha-1}\sigma^2 \\ &\to \begin{cases} \infty & \text{if } \alpha > \frac{1}{2} \\ \sigma^2 & \text{if } \alpha = \frac{1}{2} \\ 0 & \text{if } \alpha < \frac{1}{2} \end{cases} \end{aligned}$$

Central limit theorem (CLT)

Univariate CLT

- **Theorem.** $X_1, \ldots, X_n$: iid random variables:

$$\begin{aligned} &\operatorname{E}[X_i] = 0, \qquad \operatorname{Var}(X_i) = \operatorname{E}[X_i^2] = \sigma^2, \quad 0 < \sigma^2 < \infty \\ \implies \quad &\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \to_d N(0, \sigma^2) \end{aligned}$$

- CDF form, for all $t \in \mathbb{R}$:

$$\operatorname{P}\Big(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \le t\Big) \to \Phi\Big(\frac{t}{\sigma}\Big)$$

- In terms of $\bar{X}_n$:

$$\sqrt{n}\,\bar{X}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \to_d N(0, \sigma^2)$$

Non-zero mean

- $X_1, \ldots, X_n$ iid, $\operatorname{E}[X_i] = \mu \neq 0$, $\operatorname{Var}(X_i) = \sigma^2$, $0 < \sigma^2 < \infty$:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \nrightarrow_d N(\mu, \sigma^2)$$

- Add and subtract μ:

$$\begin{aligned}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i &= \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(X_i-\mu) + \frac{1}{\sqrt{n}}\,n\mu \\ &= \underbrace{\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\underbrace{(X_i-\mu)}_{\mathrm{E}[X_i-\mu]=0}}_{\to_d N(0,\sigma^2)} + \underbrace{\sqrt{n}\,\mu}_{\to\pm\infty}\end{aligned}$$

- CLT for $X_i - \mu$, for all $t \in \mathbb{R}$:

$$\begin{aligned}\mathrm{P}(\sqrt{n}(\underbrace{\bar{X}_n-\mu}_{\to_p 0}) \le t) &= \mathrm{P}\Big(\sqrt{n}\,\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu) \le t\Big) \\ &= \mathrm{P}\Big(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(X_i-\mu) \le t\Big) \\ &\to \Phi\Big(\frac{t}{\sigma}\Big)\end{aligned}$$

- Therefore:

$$\boxed{\sqrt{n}(\bar{X}_n-\mu) \to_d N(0,\sigma^2)}$$

Why $\sigma^2 > 0$

- Zero variance:

$$\begin{aligned}&\mathrm{E}[X_i] = 0, \quad \mathrm{Var}(X_i) = 0 \\ \implies\ & X_i = 0 \quad \text{with probability 1} \\ \implies\ & \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i = 0 \ \not\to_d\ N(0,\omega^2) \quad \text{for any } \omega > 0\end{aligned}$$

Proof of the CLT, step 1: moment generating function (MGF)

- MGF of X, assumed finite for t near 0:

$$M(t) = \mathrm{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx$$

- Normal MGF:

$$X \sim N(\mu,\sigma^2) \quad\iff\quad \mathrm{E}[e^{tX}] = e^{\mu t+\sigma^2 t^2/2} \ \text{ for all } t \in \mathbb{R}$$

- One can show: MGFs converge for all $t \implies$ convergence in distribution.
- Need to show, for all $t \in \mathbb{R}$:

$$\mathrm{E}\left[e^{t\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i}\right] \to e^{\sigma^2 t^2/2}$$

Proof of the CLT, step 2: product of MGFs

- Factorise, with $M(t) = \mathrm{E}\left[e^{tX_i}\right]$:

$$\begin{aligned}
\mathrm{E}\left[e^{t\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i}\right] &= \mathrm{E}\left[e^{\sum_{i=1}^n \frac{t}{\sqrt{n}} X_i}\right] \\
&= \mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_1 + \cdots + \frac{t}{\sqrt{n}}X_n}\right] \\
&= \mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_1} \times \cdots \times e^{\frac{t}{\sqrt{n}}X_n}\right] \\
&= \mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_1}\right] \times \cdots \times \mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_n}\right] \quad \text{(independence)} \\
&= \left(\mathrm{E}\left[e^{\frac{t}{\sqrt{n}}X_i}\right]\right)^n \quad \text{(identical distributions)} \\
&= \left(M\Big(\frac{t}{\sqrt{n}}\Big)\right)^n
\end{aligned}$$

Proof of the CLT, step 3: expansion of $M(t/\sqrt{n})$

- **Definition.** Small o:

$$a_n \to 0 \iff a_n = o(1), \qquad o\Big(\frac{t^2}{n}\Big) = \underbrace{\frac{t^2}{n}}_{\to 0}\, o(1)$$

- M and its derivatives at 0:

$$\begin{aligned}
M(0) &= \mathrm{E}\left[e^{tX_i}\right]\Big|_{t=0} = 1 \\
M^{(1)}(t) &= \frac{d}{dt}\mathrm{E}\left[e^{tX_i}\right] \\
&= \mathrm{E}\left[\frac{d}{dt}e^{tX_i}\right] \\
&= \mathrm{E}\left[X_i e^{tX_i}\right] \\
M^{(1)}(0) &= \mathrm{E}\left[X_i\right] = 0 \\
M^{(2)}(t) &= \mathrm{E}\left[\frac{d^2}{dt^2}e^{tX_i}\right] \\
&= \mathrm{E}\left[X_i^2 e^{tX_i}\right] \\
M^{(2)}(0) &= \mathrm{E}\left[X_i^2\right] = \sigma^2
\end{aligned}$$

- Second-order expansion around 0, then substitute the derivatives:

$$\begin{aligned}
M\Big(\frac{t}{\sqrt{n}}\Big) &= M(0) + M^{(1)}(0)\,\frac{t}{\sqrt{n}} + \frac{M^{(2)}(0)}{2}\,\frac{t^2}{n} + o\Big(\frac{t^2}{n}\Big) \\
&= 1 + 0 + \frac{\sigma^2}{2}\,\frac{t^2}{n} + \frac{t^2}{n}\,o(1) \\
&= 1 + \frac{c_n}{n}, \qquad c_n = \frac{\sigma^2 t^2}{2} + t^2\,o(1) \to \frac{\sigma^2 t^2}{2}
\end{aligned}$$

Proof of the CLT, step 4: limit of $\left(M(t/\sqrt{n})\right)^n$

- We will use, for $a_n \to \infty$:

$$\lim\left(1 + \frac{1}{a_n}\right)^{a_n} = e$$

- MGF of $\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i$, from steps 2 and 3, $t \neq 0$, $a_n = n/c_n \to \infty$:

$$\begin{aligned}
\mathrm{E}\left[e^{t\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i}\right] &= \Big(M\Big(\frac{t}{\sqrt{n}}\Big)\Big)^n \\
&= \Big(1+\frac{c_n}{n}\Big)^n \\
&= \Big(1+\frac{1}{n/c_n}\Big)^n \\
&= \Big[\underbrace{\Big(1+\frac{1}{n/c_n}\Big)^{n/c_n}}_{\to e}\Big]^{c_n} \\
&\to e^{\sigma^2 t^2/2} \quad (c_n \to \sigma^2 t^2/2, \text{ step 3})
\end{aligned}$$

- $t = 0$: both sides equal 1.
- $e^{\sigma^2 t^2/2}$: MGF of $N(0, \sigma^2)$ (step 1).
- Therefore:

$$\boxed{\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i \to_d N(0,\sigma^2)}$$

Limit $(1 + 1/n)^n \to e$

- l'Hôpital's rule in $1/n$:

$$\begin{aligned}
\lim \frac{\ln(1+1/n)}{1/n} &= \lim \frac{1/(1+1/n)}{1} \\
&= 1
\end{aligned}$$

- Take logarithms:

$$\begin{aligned}
\ln\Big[\Big(1+\frac{1}{n}\Big)^n\Big] &= n\ln\Big(1+\frac{1}{n}\Big) \\
&= \frac{\ln(1+1/n)}{1/n} \\
&\to 1 \\
\implies \quad \Big(1+\frac{1}{n}\Big)^n &= e^{\ln[(1+1/n)^n]} \\
&\to e^1 = e
\end{aligned}$$

- Same steps, $1/a_n \to 0$ in place of $1/n$: $(1 + 1/a_n)^{a_n} \to e$.

Multivariate CLT

- **Theorem.** $X_1, \ldots, X_n$: iid random $k \times 1$ vectors:

$$\begin{aligned}
&\mathrm{E}[X_i] = 0, \qquad \underset{k\times k}{\Sigma} = \mathrm{E}\left[X_i X_i^\top\right] \text{ finite and p.d.} \\
\implies \quad &\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i \to_d N(0,\Sigma)
\end{aligned}$$

- Recall: linear map of a normal vector; $a \in \mathbb{R}^k$, $a \neq 0$:

$$\underset{k\times 1}{W} \sim N(0,\Sigma) \quad \implies \quad \underset{1\times 1}{a^\top W} \sim N(0, \underset{1\times 1}{a^\top \Sigma a})$$

- Equivalent statements (one can show):

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \to_d N(0, \Sigma) \\ \iff \quad & a^\top \Big(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \Big) \to_d N(0, a^\top \Sigma a) \quad \text{for all } a \neq 0 \\ \iff \quad & \mathrm{P}\Big(a^\top \Big(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \Big) \leq t \Big) \to \Phi\Big(\frac{t}{\sqrt{a^\top \Sigma a}} \Big) \quad \text{for all } a \neq 0,\ t \in \mathbb{R} \end{aligned}$$

Proof of the multivariate CLT

- Σ p.d.:

$$a^\top \Sigma a > 0 \quad \text{for all } a \neq 0$$

- Linear combination with a:

$$a^\top \Big(\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \Big) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \underbrace{a^\top X_i}_{1 \times 1}$$

- $a^\top X_1, \ldots, a^\top X_n$: iid.
- Recall mean and variance with $\Gamma = a^\top$, $\mathrm{Var}\left(X_i\right) = \mathrm{E}\left[X_i X_i^\top\right] = \Sigma$:

$$\begin{aligned} \mathrm{E}\left[a^\top X_i\right] &= a^\top \mathrm{E}\left[X_i\right] = 0 \\ \mathrm{Var}\left(a^\top X_i\right) &= a^\top \mathrm{Var}\left(X_i\right) a = a^\top \Sigma a \end{aligned}$$

- Univariate CLT, $0 < a^\top \Sigma a < \infty$:

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^{n} a^\top X_i \to_d N(0, a^\top \Sigma a) \quad \text{for all } a \neq 0 \\ \implies \quad & \boxed{\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \to_d N(0, \Sigma)} \quad \text{(equivalent statements)} \end{aligned}$$

Summary

Convergence in probability and the LLN

- $a_n \to a$: for all $\varepsilon > 0$, some N_ε with $|a_n - a| < \varepsilon$ for all $n \geq N_\varepsilon$.
- r-th mean, $r > 0$: $\mathrm{E}\left[|\hat{\theta}_n - \theta|^r\right] \to 0$; $r = 2$: MSE.
- $\hat{\theta}_n \to_p \theta$: $\mathrm{P}(|\hat{\theta}_n - \theta| > \varepsilon) \to 0$ for all $\varepsilon > 0$.
- Markov, $M > 0$: $\mathrm{P}(|X| \geq M) \leq \mathrm{E}\left[|X|\right] / M$; Chebyshev, $\varepsilon > 0$: $\mathrm{P}(|Y - \mathrm{E}\left[Y\right]| > \varepsilon) \leq \mathrm{Var}\left(Y\right) / \varepsilon^2$.
- r-th mean $\implies \to_p$; $\mathrm{E}\left[\hat{\theta}_n\right] \to \theta$, $\mathrm{Var}\left(\hat{\theta}_n\right) \to 0 \implies \hat{\theta}_n \to_p \theta$.
- LLN: uncorrelated, $\mathrm{E}\left[X_i\right] = \mu$, $\mathrm{Var}\left(X_i\right) = \sigma^2 < \infty \implies \mathrm{Var}\left(\bar{X}_n\right) = \sigma^2 / n$, $\bar{X}_n \to_p \mu$.
- $Y_i = X_i - \mathrm{E}\left[X_i\right]$, $\mathrm{Var}\left(Y_i\right) \leq \bar{\sigma}^2$ not depending on n, $\sum_i \sum_{j \neq i} |\mathrm{Cov}\left(Y_i, Y_j\right)|$ of smaller order than $n^2 \implies \frac{1}{n} \sum Y_i \to_p 0$.
- Weak LLN: iid, $\mathrm{E}\left[|X_1|\right] < \infty \implies \bar{X}_n \to_p \mathrm{E}\left[X_1\right]$.

Vectors and Slutsky's theorem

- Vectors: $\hat{\theta}_{n,j} \to_p \theta_j$ for all $j \iff \|\hat{\theta}_n - \theta\| \to_p 0$.
- Slutsky: $\hat{\theta}_n \to_p \theta$, h continuous at $\theta \implies h(\hat{\theta}_n) \to_p h(\theta)$.
- X_i iid, $\mathrm{E}\left[X_i X_i^\top\right]$ finite and p.d. $\implies \left(\frac{1}{n}\sum X_i X_i^\top\right)^{-1} \to_p \left(\mathrm{E}\left[X_i X_i^\top\right]\right)^{-1}$.

Convergence in distribution and the CLT

- $\sqrt{n}(\hat{\theta}_n - \theta) \to_d N(0, \omega^2)$, $\omega > 0$: $\mathrm{P}(\sqrt{n}(\hat{\theta}_n - \theta) \le t) \to \Phi(t/\omega)$ for all t.
- X_i iid, mean θ, variance $0 < \sigma^2 < \infty \implies \mathrm{Var}\left(n^\alpha(\bar{X}_n - \theta)\right) = n^{2\alpha-1}\sigma^2$; $\alpha = \frac{1}{2}$: variance σ^2.
- CLT: iid, $\mathrm{E}\left[X_i\right] = 0$, $\mathrm{Var}\left(X_i\right) = \sigma^2$, $0 < \sigma^2 < \infty \implies \frac{1}{\sqrt{n}}\sum X_i \to_d N(0, \sigma^2)$.
- $\mathrm{E}\left[X_i\right] = \mu$, other conditions the same $\implies \sqrt{n}(\bar{X}_n - \mu) \to_d N(0, \sigma^2)$.
- Proof of the CLT, $\mathrm{E}\left[X_i\right] = 0$, MGF $M(t) = \mathrm{E}\left[e^{tX_i}\right]$ finite near 0: $\left(M(t/\sqrt{n})\right)^n = \left(1 + c_n/n\right)^n \to e^{\sigma^2 t^2/2}$, $c_n \to \sigma^2 t^2/2$.
- Multivariate CLT: iid $k \times 1$, $\mathrm{E}\left[X_i\right] = 0$, $\Sigma = \mathrm{E}\left[X_i X_i^\top\right]$ finite and p.d. $\implies \frac{1}{\sqrt{n}}\sum X_i \to_d N(0, \Sigma)$.
- Equivalent statements: $\frac{1}{\sqrt{n}}\sum X_i \to_d N(0, \Sigma) \iff \frac{1}{\sqrt{n}}\sum a^\top X_i \to_d N(0, a^\top \Sigma a)$ for all $a \neq 0$.